

Neeraj Singh Guleria

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SUMMARY

Data Engineer with 2.5+ years of experience designing and owning production-grade data pipelines at scale. At Optimus Electronics, I built the full pipeline stack from scratch — ADF ingestion, real-time and batch Kafka event streaming, PySpark transformations in Databricks, Delta Lake medallion layers, Snowflake migration, and Power BI delivery to senior leadership. That pipeline processes 1M+ records monthly, cut analysis time by 90%, and recovered 100+ engineering hours a month. My stack is Azure-first — ADF, Databricks, Synapse, Delta Lake, dbt, Great Expectations, Kafka, Snowflake, and Terraform — with principles that translate directly across cloud platforms. I take data quality and governance seriously enough to build the guardrails proactively — schema validation, freshness checks, RBAC, and lineage tracking are part of how I work, not optional extras.

WORK EXPERIENCE

Data Engineer — Optimus Electronics Ltd., Noida, India

March 2024 – Present

- **Cut production analysis time by 90%** by replacing shift-level manual log reviews with an automated ADF → Databricks → Synapse pipeline that processes 1M+ records every month overnight — what took engineers 3–4 hours per shift now requires zero human effort.
- **Saved 100+ engineering hours every month** by automating yield, OEE, and defect reporting across SMT, Final Assembly, and Packaging lines — eliminating per-line manual data pulls and standardising KPI reporting for production heads and senior management.
- **Built real-time event streaming pipelines using Apache Kafka** to ingest machine telemetry and production log events from MES systems — enabling sub-minute latency for defect detection alerts across all manufacturing lines, replacing batch-only ingestion cycles.
- **Designed and implemented a Kafka-based log ingestion pipeline** consuming high-volume unstructured machine logs from SMT and Final Assembly lines — events are streamed into Azure Data Lake via Kafka consumers written in PySpark (Spark Structured Streaming), then processed through the medallion pipeline.
- **Architected the full Bronze → Silver → Gold medallion pipeline from scratch** on Azure Data Lake Storage Gen2 and Delta Lake — Bronze for raw landing, Silver for PySpark cleansing and enrichment, Gold for star-schema aggregations — serving defect tracking, yield analysis, and OEE monitoring across all manufacturing lines.
- **Migrated Gold-layer analytical datasets to Snowflake** enabling cross-team self-serve analytics on a cloud-native warehouse — migrated star-schema fact and dimension tables from Synapse to Snowflake, configured Snowflake schemas and role-based access control, and validated data parity between both environments.
- **Reduced pipeline reprocessing overhead to near zero** by switching from full-load ADF runs to incremental pipelines using watermarking and tumbling window triggers — only net-new records are processed each cycle.
- **Eliminated untested SQL transformations** by introducing dbt across all Gold-layer logic — every model now has schema tests and source freshness checks; no transformation ships to production without passing them first.
- **Enforced data quality at the pipeline gate** using Great Expectations — null checks, schema assertions, and business-rule validations run before data is promoted to Silver or Gold, so broken data never reaches dashboards.
- **Made all infrastructure reproducible** by bringing ADF pipelines, Databricks clusters, Synapse pools, Kafka topics, and Key Vault secrets under Terraform — dev, staging, and production environments are now identical and version-controlled.
- **Reduced pipeline failure detection from hours to 2 minutes** by building Grafana + Azure Monitor observability with automated alerting — on-call engineers receive failure context immediately, eliminating manual log hunting.
- **Designed a star-schema Synapse SQL dedicated pool** with Fact Production, Fact Defects, Dim Machine, and Dim Product tables — the data model backing all 4 live Power BI dashboards used daily by production heads and senior management.
- **Secured the full data layer** with Azure Key Vault for all credentials, RBAC policies for role-level access control, and Unity Catalog for data lineage tracking and metadata management across Databricks.
- **Halved MES extraction query time** by pushing complex filtering logic into PL/SQL stored procedures at the source database rather than pulling raw data and filtering in Spark — reducing ingestion overhead significantly.

Data Analyst Intern — Living Things, Mumbai, India

October 2023 – January 2024

- **Automated weekly data validation workflows** using Python and SQL — standardised pipelines replaced manual Excel-based checks, reducing report preparation time for the operations team.
- **Surfaced operational insights from business datasets** that fed directly into leadership planning decisions — findings were communicated through Python visualisations to non-technical stakeholders.

- **Configured and tested a GPON network** for an IPMPLS Defense Project, supporting IP address planning, hardware setup, and network troubleshooting.
- **Gained hands-on exposure** to network operations, TCP/IP protocols, and network security fundamentals.

SKILLS

Cloud Platforms: Microsoft Azure, Azure Data Factory (ADF), Azure Databricks, Azure Synapse Analytics, ADLS Gen2, Azure Event Hubs, Azure Cosmos DB, Azure SQL Database, Azure Key Vault, Azure Purview, Snowflake
Big Data & Streaming: Apache Spark, PySpark, Spark Structured Streaming, Apache Kafka, Delta Lake, Apache Airflow

ETL & Data Engineering: dbt, ETL/ELT Pipeline Design, Medallion Architecture, Incremental Loading, Watermarking, SCD Type 2, Metadata-Driven Pipelines, Data Vault 2.0, REST API Integration

Databases & Warehousing: SQL (Advanced), PL/SQL, Azure Synapse Dedicated SQL Pools, Snowflake, Oracle, MySQL, Star Schema, Snowflake Schema

Programming: Python, PySpark, Pandas, NumPy, SQL, PowerShell, Bash, C#, JavaScript

Data Modelling & Governance: Star Schema, Snowflake Schema, Data Vault 2.0, RBAC, Data Lineage, Unity Catalog, Azure Purview

DevOps & CI/CD: GitHub Actions, Terraform, Azure DevOps, ARM Templates, Bicep, Docker

BI & Monitoring: Power BI, Azure Monitor, Grafana, Log Analytics

Data Formats: Parquet, Avro, JSON, Delta Lake

PROJECTS

Azure Manufacturing Data Pipeline

Problem: Manual production log analysis took several hours per cycle, creating delays in defect detection and yield reporting for 1 million+ monthly records.

Solution: Built a complete ETL pipeline using Azure Data Factory, Apache Kafka (real-time machine event streaming), Azure Databricks (PySpark + Spark Structured Streaming), and Azure Synapse Analytics with medallion architecture, Delta Lake optimisation, Great Expectations data validation, and Azure Monitor alerting.

Outcome: Reduced analysis time by 90%, enabled real-time Power BI dashboards, and fully automated reporting — recovering 100+ engineering hours per month.

Snowflake Data Migration & Self-Serve Analytics

Problem: Synapse-based Gold layer created dependency on a single warehouse platform, limiting cross-team analytics access and scalability.

Solution: Migrated star-schema fact and dimension tables from Azure Synapse to Snowflake — configured Snowflake schemas, role-based access control, and data pipelines; validated full data parity between environments using automated reconciliation scripts in Python.

Outcome: Enabled cross-team self-serve analytics on Snowflake with proper governance, reducing data access bottlenecks and providing a cloud-native warehouse layer alongside existing Synapse infrastructure.

MES Web Application for Self-Serve Production Reporting

Problem: Production teams depended on IT to run complex SQL queries for historical data, creating bottlenecks and delays in operational decisions.

Solution: Developed an ASP.NET Core MVC web application with dynamic filters, server-side pagination, and optimised PL/SQL stored procedures for fast data retrieval.

Outcome: Reduced manual query turnaround time by 80%, enabling production teams to self-serve real-time reports without IT involvement.

End-to-End Azure Data Engineering Pipeline

Problem: Organisation lacked scalable data infrastructure for business intelligence reporting and executive KPI tracking.

Solution: Designed and implemented full medallion architecture using ADF, Azure Databricks, Synapse Analytics, and Power BI, including incremental loads, SCD Type 2 handling, and star schema modelling.

Outcome: Delivered 3 executive dashboards backed by 5 fact tables and 12 dimension tables, improving data accuracy, consistency, and reporting speed across teams.

EDUCATION

Bachelor of Science (Hons.) – Data Science

2020 – 2023

K.R. Mangalam University, Gurgaon, Haryana

Relevant coursework: Statistical Data Analysis, Python, SQL, Data Mining, Machine Learning, Power BI, DBMS

CERTIFICATIONS

- Azure Data Engineering Masters: Build Scalable Solutions – [Udemy](#) (2025)
- Azure Data Engineering for Data Engineers: DP-203 and Beyond – [Udemy](#) (2026)